

# assignment nomentum II

 This is a preview of the published version of the quiz

Started: Apr 29 at 8:02pm

## Quiz Instructions

2 attempts



Question 1 14.5 pts

A 0.10-kg ball, traveling horizontally at 25 m/s, strikes a wall and rebounds at 19 m/s. What is the magnitude of the change in the momentum of the ball during the rebound?

☐

4.4 kg · m/s

☐

5.4 kg · m/s

☐

1.2 kg · m/s

☐

72 kg · m/s

☐

1.8 kg · m/s



Question 2 14.5 pts

A 0.140-kg baseball is thrown with a velocity of  27.1 m/s. It is struck by the bat with an average force of 5000 N, which results in a velocity of 37.0 m/s in the opposite direction from the original velocity. How long were the bat and ball in contact?

☐

$4.30 \times 10^{-3} \text{ s}$

☐

$1.79 \times 10^{-3} \text{ s}$

☐

$1.28 \times 10^{-2} \text{ s}$

☐

$3.07 \times 10^{-2} \text{ s}$



Question 3 14.5 pts

A 1200-kg car moving at 15.6 m/s suddenly collides with a stationary car of mass 1500 kg. If the two vehicles lock together, what is their combined velocity immediately after the collision?

☐

5.5 m/s

☐

6.9 m/s

☐

8.6 m/s

☐

12.1 m/s

☒

Question 4 14.5 pts

A 1000-kg whale swims horizontally to the right at a speed of 6.0 m/s. It suddenly collides directly with a stationary seal of mass 200 kg. The seal grabs onto the whale and holds fast. What is the speed of these two sea creatures just after their collision? You can neglect any drag effects of the water during the collision.

☐

3.0 m/s

☐

6.0 m/s

☐

5.0 m/s

☐

0.00 m/s

☐

4.0 m/s

☒

Question 5 14.5 pts

A 328-kg car moving at 19.1 m/s in the +x direction hits from behind a second car moving at 13 m/s in the same direction. If the second car has a mass of 790 kg and a speed of 15.1 m/s right after the collision, what is the velocity of the first car after this sudden collision?

hint: Before they go in the same direction. Use  $m_1V_1 + m_2V_2 = m_1V_1' + m_2V_2'$

☐

14.0 m/s

☐

24.2 m/s

☐

18.2 m/s



-14.0 m/s



Question 6 14.5 pts

A golf club exerts an average horizontal force of 1000 N on a 0.045-kg golf ball that is initially at rest on the tee. The club is in contact with the ball for 1.8 ms. What is the speed of the golf ball just as it leaves the tee?



45 m/s



35 m/s



50 m/s



40 m/s



30 m/s



Question 7 16 pts

A 1200-kg ore cart is rolling at 10.8 m/s across a flat friction-free surface. A crane suddenly drops 858kg of ore vertically into the cart. How fast does the cart move just after being loaded with the ore?

hint: its a collision hit and stick. But along the horizontal the load is not moving  
so  $m_1 V_1 + 0 = (m_1 + m_2) V$



4.20 m/s



3.80 m/s



5.70 m/s



6.30 m/s

Not saved

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